

Tarea Decisión 4.7

36p 7302

3) $a = ?$ $ab = 100$
 $b = ?$ $a = 100/b$

$$S = a + b$$

$$S = \left(\frac{100}{b}\right) + b = \frac{100 + b^2}{b}$$

$$S' = \frac{2b(b) - (100 + b^2)}{b^2} = \frac{b^2 - 100}{b^2} = 0$$

$$b^2 - 100 = 0$$

$$b = \pm\sqrt{100}$$

$$b = 10$$

	$f'(x)$
0, 10	-
10	
10, 20	+

$$a = \frac{100}{b}$$

$$a = \frac{100}{10}$$

$$a = 10$$

5) $y = x + 2$
 $y = x^2$

$$a = x + 2 + x^2$$

$$a = x^2 + x + 2$$

$$a' = 2x + 1 = 0$$

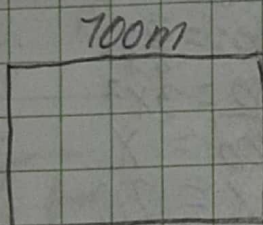
$$2x = -1$$

$$x = -\frac{1}{2}$$

	$f'(x)$
-1, -1/2	-
-1/2	
-1/2, 2	+

$$7/4$$

7)



$$100 = 2a + 2b$$

$$100 - 2b = 2a$$

$$\frac{100 - 2b}{2} = a$$

$$50 - b = a$$

$$50 - 25 = a$$

$$a = 25m$$

$$A = ab$$

$$A = (50 - b)b$$

$$A = 50b - b^2$$

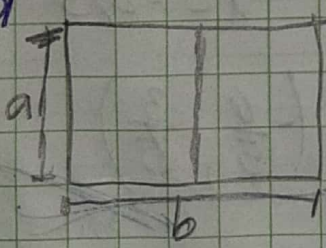
$$A' = 50 - 2b = 0$$

$$2b = 50$$

$$b = 25m$$

$f'(x)$	$f'(x)$
0, 25	+
25	
25, 100	-

13)



$$A = ba = 15,000m^2$$

$$a = \frac{15,000}{b}$$

$$L = 2b + 3a$$

$$L = 2b + 3\left(\frac{15,000}{b}\right)$$

$$L = 2b + \frac{45,000}{b}$$

Scribe

$$L' = 0.6 - \frac{45,000}{b^2} = 2 - \frac{45,000}{b^2} = 0$$

	$f'(x)$	$f(x)$
0,150	-	
150		
150,15,000	+	

$$-\frac{45,000}{b^2} = -2$$

$$45,000 = 2b^2$$

$$22,500 = b^2$$

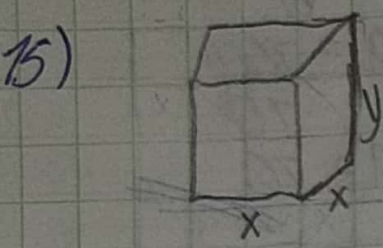
$$\sqrt{22,500} = b$$

$$b = 150m$$

$$a = \frac{75,000}{b}$$

$$a = \frac{75,000}{150}$$

$$a = 500m$$



15)

$$A = 2x^2 + 4xy$$

$$V = x^2 y$$

$$1200 = x^2 + 4xy$$

$$1200 - x^2 = 4xy$$

$$\frac{1200 - x^2}{4x} = y$$

	$F'(x)$
0,20	+
20	
20,1200	-

$$\frac{1200 - (20)^2}{4(20)} = y$$

$$y = 70$$

$$V = \frac{1200x^2 - x^3}{4x}$$

$$V = 300x - \frac{1}{4}x^3$$

$$V' = 300 - \frac{3}{4}x^2 = 0$$

$$300 = \frac{3}{4}x^2$$

$$1200 = 3x^2$$

$$\sqrt{400} = x$$

$$x = 20$$

$V = x^2 y$
 $V = (20)^2 (70)$
 $V = 4,000 \text{ cm}^3$ es el mayor volumen posible de la caja

21) $y = 2x + 3$

$P = (0,0)$

$$-3y = 2x$$

$$\frac{-3y}{2} = x$$

$$x = -3 \left(\frac{3}{2} \right)$$

$$x = -\frac{9}{2}$$

$$d = \sqrt{(x-0)^2 + (y-0)^2}$$

$$d = \sqrt{\left(-\frac{3y}{2}\right)^2 + (y-0)^2}$$

$$d' = \frac{1 \cdot \left(-\frac{3y}{2}\right) \cdot \left(-\frac{3}{2}\right) + 2y \cdot 1}{2 \sqrt{\frac{9y^2}{4} + y^2}} = 0$$

$$\left(-\frac{9}{5}, \frac{3}{5}\right)$$

$$\frac{y-3}{2} - y = 0$$

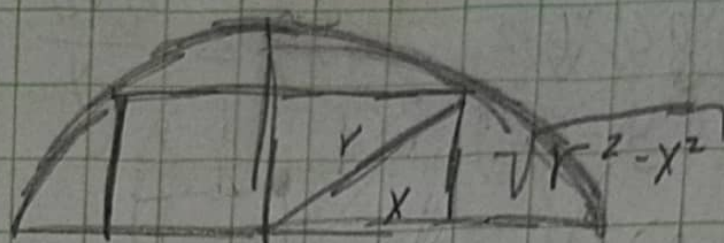
$$\frac{y-3+4y}{2} = 0$$

$$5y-3=0$$

$$5y=3$$

$$y = \frac{3}{5}$$

25)



$$A = x \sqrt{r^2 - x^2}$$

$$A^2 = x^2(r^2 - x^2) = r^2x - x^4$$

$$A' = 2r^2x - 4x^3 = 0$$

$$2x(r^2 - 2x^2) = 0$$

$$A\left(\frac{r}{\sqrt{2}}\right) = \frac{r}{\sqrt{2}} \sqrt{r^2 - \left(\frac{r}{\sqrt{2}}\right)^2}$$

$$\frac{r}{\sqrt{2}} \frac{r}{\sqrt{2}} = \frac{r^2}{2}$$

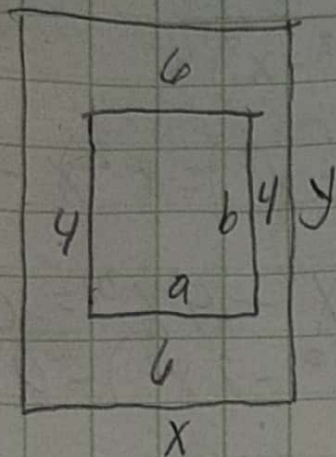
$$x = 0$$

$$r^2 - 2x^2 = 0$$

$$x = \frac{r}{\sqrt{2}}$$

Dimensiones del rectángulo $x = \frac{2}{\sqrt{2}}r = \sqrt{2}r$

35)



	$f'(x)$	$f(x)$
0,24	-	
24		
24,0	+	

$$x = \frac{384}{y}$$

$$x = \frac{384}{24} =$$

$$x = 16$$

$$A = xy = 384 \text{ cm}^2$$

$$A = (x-8)(y-12)$$

$$x = \frac{384}{y}$$

$$A = \left(\frac{384}{y} - 8\right)(y - 12)$$

$$A = \left(\frac{384 - 8y}{y}\right)(y - 12)$$

$$A = \frac{384y - 4.608 - 8y^2 + 96y}{y}$$

$$A = \frac{480y - 4.608 - 8y^2}{y} = \frac{-8y^2 + 480y - 4.608}{y}$$

$$A' = \frac{(-16y + 480)(y) - (-8y^2 + 480y - 4.608)(1)}{y^2}$$

$$A' = \frac{-16y^2 + 480y + 8y^2 - 480y + 4.608}{y^2}$$

$$A' = \frac{-8y^2 + 4.608}{y^2} = 0$$

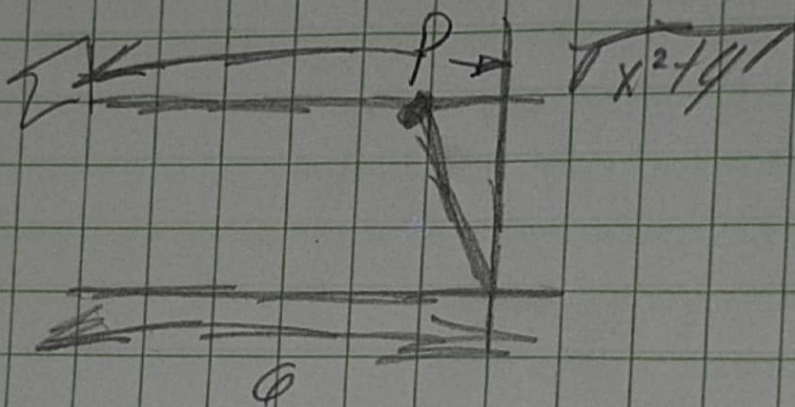
$$4.608 = 8y^2$$

$$4.608/8 = y^2$$

$$\sqrt{576} = y$$

$$y = 24$$

51)



$$CT(x) = (6-x)4 + (\sqrt{x^2+4})8$$

$$CT'(x) = -4 + 8 \cdot \frac{1}{2} (x^2+4)^{-1/2} (2x)$$

$$CT'(x) = -4 + \frac{8x}{\sqrt{x^2+4}} = 0 \Rightarrow \frac{8x}{\sqrt{x^2+4}} = 4 \Rightarrow \sqrt{x^2+4} = 2x$$

$$CT(x) = \sqrt{x^2+4} = (2x)^2$$

$$CT(x) = x^2 + 4 = 4x^2$$

$$3x^2 = 4$$

$$x = \sqrt{\frac{4}{3}} = \frac{2}{\sqrt{3}}$$

$$F(x) = 6 - x$$

$$F\left(\frac{2}{\sqrt{3}}\right) = 6 - \frac{2}{\sqrt{3}} = 4.85 \text{ kN}$$